
When and How to Canonize: a Generalization Perspective

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Abstract

1 While equivariant architectures are standard for processing symmetric data, there
2 is growing interest in achieving equivariance by applying group averaging or canon-
3 ization to non-equivariant backbones. However, the theoretical generalization
4 properties of these alternative strategies remain poorly understood. We introduce
5 a theoretical framework to analyze the generalization error of these methods by
6 bounding their covering numbers. We establish a rigorous generalization hierarchy:
7 the error bounds of canonized models are at best equal to the error bounds of struc-
8 turally equivariant and group-averaged models, and at worst equal to the bounds of
9 non-equivariant baselines. Furthermore, we show that there exist "optimal" canon-
10 izations which attain the optimal error bounds, and "poor" canonizations which
11 attain the non-equivariant error bounds, and that this depends on the regularity
12 of the canonization. Finally, applying this framework to permutation groups in
13 point cloud processing, we rigorously prove that the covering number of lexico-
14 graphical sorting grows exponentially with point cloud dimension, whereas Hilbert
15 curve canonization guarantees polynomial growth. This provides the first formal
16 theoretical justification for the empirical success of Hilbert curve serialization
17 in state-of-the-art point cloud architectures. We conclude with experiments that
18 support our theoretical claims.

19 1 Introduction

20 The integration of geometric priors into neural networks has become a cornerstone of modern machine
21 learning, particularly for applications involving graphs, 3D point clouds, and molecular structures. In
22 these domains, the underlying data often exhibits inherent symmetries, meaning the target function to
23 be learned is equivariant to a specific group action. Exploiting these symmetries is known to improve
24 sample complexity and generalization [8, 5].

25 While specialized architectures, which are equivariant by design, are a common method for pro-
26 cessing symmetric data, there is growing interest in achieving equivariance by applying generalized
27 group averaging methods to non-equivariant backbones. This family of methods includes full group
28 averaging, which is theoretically sound but intractable for all but very small groups, group augmen-
29 tation, which can be seen as an efficient approximation of group averaging, frame averaging [22],
30 which enables equivariant averaging over subsets of the group, and canonization, which maps every
31 group orbit to a single, consistent "canonical" element. Canonization is the most efficient of all the
32 generalized group averaging, as it only involves processing a single element. At the same time, it
33 enjoys the same universal approximation guarantees as full group averaging [14]. These observations
34 have motivated researchers to define canonizations in several diverse domains. These include sign
35 canonizations for spectral embeddings of graphs [19, 18, 12], canonizations of graphs [17] and point
36 clouds [14, 2, 10, 33] including the celebrated Point Transformer V3 [29], and canonization of the
37 SMILES representation of small molecules [28].

38 In this paper, we provide a theoretical study of canonization from the perspective of generalization.
 39 Our focus is both on comparing canonization with other options for enforcing equivariance and on
 40 comparing the quality of different canonizations. Our main contributions are threefold:

- 41 1. **Canonization vs. Group Averaging:** We prove that the generalization bounds of canonized
 42 models are at best equal to the bounds of group-averaged models, and at worst equal to the
 43 bounds of non-equivariant models.
- 44 2. **Canonizations and continuity:** We show that continuous, isometric canonizations attain the
 45 *optimal* generalization error achieved by group averaging, while discontinuous canonizations
 46 can lead to *poor* canonizations whose error bounds are identical to non-equivariant models.
- 47 3. **Point Clouds Serialization:** We prove that canonizing point clouds with respect to per-
 48 mutation via lexicographical sorting is inferior to the Hilbert canonizations used in Point
 49 Transformer V3 [29], thus establishing a theoretical justification for the empirical success of
 50 this method.

51 1.1 Related Work

52 **Canonization vs. Group Averaging** It was observed that randomized SMILES can often outper-
 53 form canonized SMILES [1, 3] (see also [13]). Our first conclusion gives a theoretical justification
 54 for this empirical observation.

55 **Discontinuity** It was shown in [7, 2] that in many equivariant learning scenarios, continuous
 56 canonizations are mathematically impossible. Attempts to mitigate this issue were made in [26, 16].
 57 Our second conclusion complements these works by showing how discontinuity of canonization hurts
 58 the generalization of the canonized model.

59 **Equivariance and Generalization** Many papers have considered generalization analysis of equiv-
 60 ariant models [27, 21, 20, 9], but they do not consider canonizations. To our knowledge, the only
 61 paper considering the generalization of canonizations is [25]. They prove that in some setting the
 62 *approximation error* of canonized models is lower than group averaged models, and as a result the
 63 *expected error* of canonized models can be lower when the model is presented with enough samples.
 64 In contrast, our first conclusion is that the *generalization error* of group average models is always
 65 lower than the generalization error of canonized models.

66 1.2 Notation

67 Throughout the paper, we will consider a metric space (K, ρ) , where K is endowed by the action of a
 68 group $\mathbb{G}$. We will assume that the metric is $\mathbb{G}$ invariant, i.e., $\rho(g \cdot x, g \cdot y) = \rho(x, y)$ for all $x, y \in K$
 69 and $g \in \mathbb{G}$.

70 **Quotient space** We denote the orbit of $x \in K$ by $[x] = \{gx \mid g \in G\}$, and denote the quo-
 71 tient space to be the space of orbits $K/\mathbb{G} := \{[x] \mid x \in K\}$. On this space, we define a metric
 72 $\rho_{\mathbb{G}}([x], [y]) := \min_{g \in \mathbb{G}} \rho(g \cdot x, y)$. The $\mathbb{G}$ invariance of ρ guarantees that $\rho_{\mathbb{G}}$ is symmetric and
 73 satisfies the triangle inequality. To ensure that $\rho_{\mathbb{G}}([x], [y]) = 0$ implies that $[x] = [y]$, we explicitly
 74 add the assumption that the minimum in the definition of $\rho_{\mathbb{G}}$ always exists. A triplet $(K, \rho, \mathbb{G})$
 75 satisfying the conditions of the last two paragraphs is called a *module*, and all examples considered in
 76 this paper are modules.

77 **Covering number** Let (K, ρ) be a metric space. We say that $C \subset K$ is an ϵ cover of K , if
 78 $\forall x \in K, \exists y \in C : \rho(x, y) \leq \epsilon$. The minimal N such that there exists an ϵ cover of cardinality N is
 79 called the ϵ covering number of K , and is denoted by $\mathcal{N}(K, \rho, \epsilon)$.

80 **Canonization** We say that $c : K \rightarrow K$ is a canonization if for every $x \in K$, (i) x is mapped to an
 81 element in its orbit, namely $c(x) \in [x]$, and (ii) all elements in the orbit $[x]$ are mapped to the same
 82 element.

83 2 Generalization, Canonization and Group Averaging

Consider a supervised learning problem of learning an unknown function $f : X \rightarrow Y$ from a finite set of n samples $S = \{(x_i, f(x_i)) \mid i = 1, \dots, n\}$ of the function. Assume that we have some algorithm (e.g., gradient descent) which, given S , returns a hypothesis $h_S : X \rightarrow Y$. Let $\ell : Y \times Y \rightarrow \mathbb{R}$ be a function which will be used as a loss function. The *expected loss* $\mathcal{L}$ and the *empirical loss* $\mathcal{L}_{\text{emp}}$ are defined via

$$\mathcal{L}_{\text{exp}}(h_S) \triangleq \mathbb{E}_{x \sim \mu} \ell(h_S(x), f(x)), \quad \mathcal{L}_{\text{emp}}(h_S) \triangleq \frac{1}{n} \sum_{i=1}^n \ell(h_S(x_i), f(x_i))$$

84 The *generalization error* of h_S is the absolute difference between the expected and empirical loss.
 85 The generalization error of h_S can be bounded using the Lipschitz constants of the function h_S, f, ℓ
 86 and the covering number of the domain X , as follows:

Theorem 2.1. *Let (X, ρ_X) and (Y, ρ_Y) be metric spaces and assume X is compact. Let $f : X \rightarrow Y$ be a c_f Lipschitz function, and let $\ell : Y \times Y \rightarrow [0, M]$ be a c_ℓ Lipschitz function. Then for any $\epsilon, \delta > 0$ and natural n , for a training sample set S generated by n IID draws from a distribution μ , with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{\text{exp}}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M \sqrt{\frac{2\mathcal{N}(X, \rho_X, \epsilon) \ln 2 + 2 \ln(1/\delta)}{n}}$$

87 where c_{h_S} denotes the Lipschitz constant of h_S .

88 This theorem follows easily from a well-known theorem from [31], which is often employed in the
 89 study of neural network generalization [27]. The derivation from the Theorem is given in Appendix
 90 A.1.

91 We now apply this theorem in the setting where the learned function f is invariant to a group action.
 92 In the following proposition, we consider three types of models: (i) models that do not exploit
 93 function symmetry at all, (ii) models that are invariant by construction, and (iii) non-invariant models
 94 composed with canonizations.

Proposition 2.2. *Let $(K, \rho, \mathbb{G})$ be a module. Let (Y, ρ_Y) be a metric space, and assume K is compact. Let $f : K \rightarrow Y$ be a c_f Lipschitz function **which is $\mathbb{G}$ invariant**, and let $\ell : Y \times Y \rightarrow [0, M]$ be a c_ℓ Lipschitz function, then for any $\epsilon, \delta > 0$ and natural n , for a training sample set S generated by n IID draws from the distribution μ , with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{\text{exp}}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M \sqrt{\frac{2\mathcal{N}(h_S) \ln 2 + 2 \ln(1/\delta)}{n}}$$

95 where

$$\mathcal{N}(h_S) = \begin{cases} \mathcal{N}(K, \rho, \epsilon) & \text{if } h_S \text{ is } c_{h_S} \text{ Lipschitz} \\ \mathcal{N}(c(K), \rho, \epsilon) & \text{if } h_S = \tilde{h}_S \circ c, \tilde{h}_S \text{ is } c_{\tilde{h}_S} \text{ Lipschitz, and } c \text{ is a canon.} \\ \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) & \text{if } h_S \text{ is } c_{h_S} \text{ Lipschitz and } \mathbb{G} \text{ invariant} \end{cases} \quad (1)$$

96 *Proof idea.* The claim for a Lipschitz function h_S without any invariant structure follows immediately
 97 from Theorem 2.1. For a function of the form $h_S = \tilde{h}_S \circ c$, the claim follows from Theorem 2.1
 98 when replacing the domain K with $c(K)$ and the function h_S with $\tilde{h}_S$. The claim for h_S which is
 99 both Lipschitz and invariant follows from the fact that due to invariance h_S, f can be identified with
 100 function $\hat{h}_S, \hat{f} : K/\mathbb{G} \rightarrow Y$ satisfying $\hat{h}_S([x]) = h_S(x), \hat{f}([x]) = f(x), \forall x \in K$, and the Lipschitz
 101 constant remains unchanged after the identification (see Lemma 20 in [23]). $\square$

103 For each of the three model types considered in Proposition 2.2, a different value of $\mathcal{N}(h_S)$ is
 104 provided. The following simple lemma establishes the relationship between these three values:

105 **Lemma 2.3.** *For any module $(K, \rho, \mathbb{G})$, and any $\epsilon > 0$, the following inequality holds:*

$$\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \mathcal{N}(c(K), \rho, \epsilon) \leq \mathcal{N}(K, \rho, \epsilon) \quad (2)$$

to self: add
proof for
equivariant
setting in
appendix,
mention in
main text

106 *Proof.* The inequality $\mathcal{N}(c(K), \rho, \epsilon) \leq \mathcal{N}(K, \rho, \epsilon)$ is immediate since $c(K) \subseteq K$. For the other
 107 inequality, note that the quotient mapping $q : c(K) \rightarrow K/\mathbb{G}$ defined by $q(x) = [x]$ is onto and
 108 1-Lipschitz, and therefore it transforms any ϵ cover of $c(K)$ to an ϵ cover of $K/\mathbb{G}$, thus implying that
 109 $\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \mathcal{N}(c(K), \rho, \epsilon)$.

110 □

111 Informally, the conclusion we draw from this lemma is as follows:

Conclusion 1

The generalization bounds of canonized models are at best equal to the bounds of group-averaged models, and at worst equal to the bounds of non-equivariant models.

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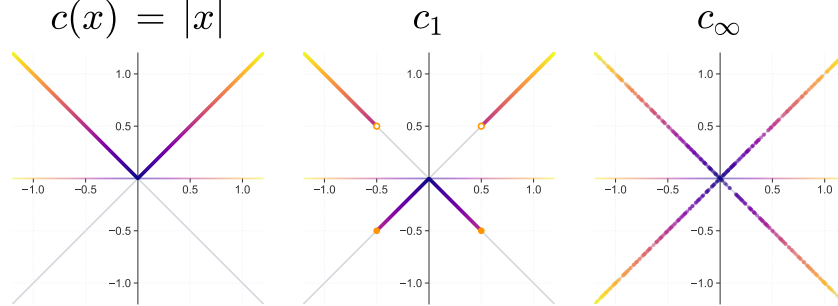
114 While this conclusion is valid for all equivariant models, it is most relevant for equivariant models
 115 obtained from group averaging a non-equivariant model. In this setting, the same backbone model
 116 can be used for canonization and group averaging, and so a direct comparison can be made.

117 We conclude this section by showing that the ratio between the largest and smallest covering numbers
 118 in (2) is bounded by the group's cardinality:

119 **Proposition 2.4.** [See proof in A.3] Let $(K, \rho, \mathbb{G})$ be a module, and assume $\mathbb{G}$ is finite. Then

$$\mathcal{N}(K, \rho, \epsilon) \leq |\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon).$$

Figure 1: Visualization of three canonizations for the action of $\mathbb{G} = \{-1, 1\}$ on $\mathbb{R}$. The natural canonization $c(x) = |x|$, which is shown to be optimal (Example 3.3), and two discontinuous canonizations, which on an appropriate domain are poor (Example 3.5).



isnt the proof done for invariant models? In general, it feels "equivariant" and "invariant" are switched in and out, which is a bit confusing. Maybe clarify?

120 3 Canonizations and Continuity

121 In the previous section, we established upper and lower bounds for the covering numbers of canoniza-
 122 tions. Our focus in this section is to study when these bounds are achieved. As we will see, this will
 123 be related to the continuity of the canonization. We begin by defining optimal and poor canonizations.

124

125 **Definition 3.1.** Let $(K, \rho, \mathbb{G})$ be a module and let ϵ be a positive number. We say that c is (K, ϵ)
 126 *optimal* if it attains the lower bound in (2), namely $\mathcal{N}(c(K), \rho, \epsilon) = \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon)$. We say that c
 127 is (K, ϵ) *poor* if it attains the upper bound in (2), namely $\mathcal{N}(c(K), \rho, \epsilon) = \mathcal{N}(K, \rho, \epsilon)$.

128 We will next show conditions for optimal and poor canonizations, and show how these conditions
 129 relate to the continuity of the canonization.

130 **Optimal canonizations:** First, we prove canonizations, which are isometries, are optimal. We say
 131 that a canonization $c : K \rightarrow K$ is an *isometry* if $\rho(c(x), c(y)) = \rho_{\mathbb{G}}([x], [y])$ for all $x, y \in K$.

132 **Claim 3.2.** [proof in A.3] Any isometric canonization $c : K \rightarrow K$ is (K, ϵ) optimal for all $\epsilon > 0$.

133 We next use this claim to give several examples of optimal canonizations:

134 **Example 3.3.** The following canonizations are isometries (see proof in A.3)

- 135 1. The canon. $c(x) = |x|$ with respect to the action of $\{-1, 1\}$ on $\mathbb{R}$ and the standard metric.
- 136 2. The canon. $\text{sort} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ with respect to the action of permutations and a metric ρ
- 137 induced by a p norm.
- 138 3. The centralization canon. defined for point clouds in $\mathbb{R}^{d \times n}$ with respect to the action of
- 139 translation by vectors in $\mathbb{R}^d$, with the metric ρ induced by the Frobenius norm.

140 **Poor canonizations:** We next show that discontinuities in canonizations can lead to poor canoniza-
 141 tions. This occurs under two conditions: firstly, the canonization is 'strongly discontinuous' in the
 142 sense that it has $|G|$ partial limits. Secondly, the domain K of the group action is chosen adversarially
 143 to target the canonization's discontinuity.

144 **Proposition 3.4.** *[proof in A.3] Let $(V, \rho, \mathbb{G})$ be a module, where $\mathbb{G}$ is a finite group. Let c be a*
 145 *canonization such that c has $|G|$ partial limits at a point $x \in V$, then for any small enough $\epsilon > 0$*
 146 *there exists a $\mathbb{G}$ invariant set $K \subseteq V$ such that $\mathcal{N}(c(K), \rho, \epsilon) = \mathcal{N}(K, \rho, \epsilon) = |\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho, \epsilon)$.*

147 Based on this proposition, we give two examples of poor canonizations:

Example 3.5. Consider the action of $\mathbb{G} = \{-1, 1\}$ on $V = \mathbb{R}$. Consider the canonizations

$$c_1(t) = \begin{cases} |t| & \text{if } |t| > 1/2 \\ -|t| & \text{if } |t| \leq 1/2 \end{cases}, \quad c_\infty(t) = \begin{cases} |t| & \text{if } t \in \mathbb{Q} \\ -|t| & \text{if } t \notin \mathbb{Q} \end{cases}.$$

148 Both of these canonizations are discontinuous and have a point with $|G| = 2$ partial limits. Therefore,
 149 by Proposition 3.4 for every small enough $\epsilon > 0$ we can choose an adversarial domain for which
 150 the canonization is poor. For c_1 we can choose for every $\epsilon \in (0, 1)$ the set $K = B_\epsilon(-\frac{1}{2}) \cup B_\epsilon(\frac{1}{2})$
 151 and then $\mathcal{N}(K, \rho, \epsilon) = \mathcal{N}(c_1(K), \rho, \epsilon) = 2$ and so the canonization is poor. In the canonization c_∞
 152 the discontinuities are much more extreme. In this case, we can choose a non-adversarial domain
 153 $K = [-1, 1]$, and the canonization will be poor for any $\epsilon > 0$ because the covering number of a set
 154 and its closure are the same, and in our case, the closure of $c_\infty(K)$ will be all of K .

155 **Example 3.6.** Consider the action of the permutation group $\mathbb{G} = S_n$ on the space of matrices $\mathbb{R}^{d \times n}$
 156 by permuting the columns of the matrices. We call such matrices 'point clouds'. We assume that
 157 $n, d \geq 2$ (the case $d = 1$ sorting is an isometry by Example 3.3). A natural canonization in this
 158 scenario is c_{lex} , which permutes the columns of the matrices so that the first row of the matrix is
 159 sorted from small to large, and in the event of ties in the first row, sorts according to the second
 160 row, and continues in this fashion until a permutation is uniquely defined. The canonization c_{lex}
 161 is discontinuous. Moreover, the canonization has $n! = |\mathbb{G}|$ partial limits at any matrix $M \in \mathbb{R}^{d \times n}$
 162 satisfying that the matrix does not contain two identical columns, but $M_{1,1} = M_{1,2} = \dots = M_{1,n}$.
 163 Thus, according to Proposition 3.4, for every small enough $\epsilon > 0$ there exists an adversarial domain
 164 for which the canonization is poor.

Conclusion 2

We show that isometric canonizations are optimal, while discontinuous canonizations can yield poor canonizations. These results give a rigorous justification for the importance of continuous canonizations.

165

166 4 Case study: Permutation Canonizations

167 In Example 3.6, we showed that lexicographical sorting is a poor canonization on an adversarial
 168 domain. In this section, we analyze the covering numbers of lexicographical sorting when considering
 169 a fixed, 'natural' domain of point clouds. We will show that even in this non-adversarial setting, the
 170 covering number of lexsort canonization is much higher than the covering number of the quotient
 171 space: we show this by showing that the covering number of lexsort grows exponentially in n , the
 172 number of points in the point cloud, while the covering number of the quotient space only grows
 173 polynomially in n . Moreover, we will consider an alternative canonization based on Hilbert curves,
 174 and show that its covering numbers also grow only polynomially in n . The significance of this result
 175 is that it gives a theoretical justification for the choice of Point Transformer V3 [29] to use Hilbert
 176 canonizations for point cloud serialization.

Throughout this section we consider the module $(K, \rho, \mathbb{G})$ with the domain $K = [0, 1]^{d \times n}$, the distance induced by the element-wise infinity norm $\rho(X, Y) = \|X - Y\|_\infty$, and the permutation group $\mathbb{G} = S_n$. We will show that for fixed ϵ, d , the covering number of c_{lex} grows exponentially in n , while the covering number of the quotient space and Hilbert canonizations only grows polynomially in n . We begin by proving the exponential growth of lexsort canonizations:

Lemma 4.1. *[proof in A.4] Let $k, d, n \geq 2$ be natural numbers, and set $\epsilon = 1/(2k)$, $K = [0, 1]^{d \times n}$ and $\mathbb{G} = S_n$. Then the covering number of the lexicographical sorting canonization satisfies*

$$\left(\frac{1}{2\epsilon}\right)^{(d-1) \cdot n + 1} \leq \mathcal{N}(c_{\text{lex}}(K), \rho, \epsilon). \quad (3)$$

Proof idea. The lexsort canonization is ineffective on the set of its 'severe discontinuities' - point clouds whose first row is constant. These point clouds form a $(d-1) \cdot n + 1$ dimensional hypercube. The covering number of $c_{\text{lex}}(K)$ is upper bounded by the covering number of this hypercube, which is the left-hand side of (3). $\square$

Next, we prove the polynomial growth of the quotient space's covering number:

Lemma 4.2. *Let n, d be natural numbers and $\epsilon \in (0, 1)$. Set $K = [0, 1]^{d \times n}$ and $\mathbb{G} = S_n$. Then the covering number of the quotient space satisfies*

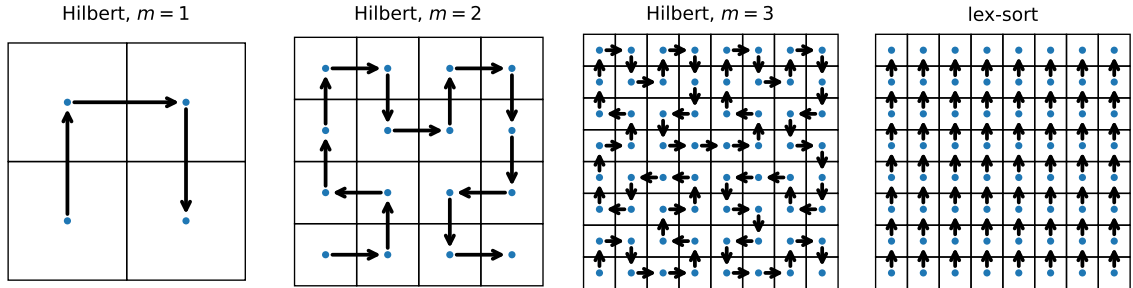
$$\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \binom{n + k^d - 1}{n}, \text{ where } k = \left\lceil \frac{1}{2\epsilon} \right\rceil. \quad (4)$$

Proof. Set $k = \lceil 1/(2\epsilon) \rceil$. The set K can be covered by k^d hypercubes of diameter k and radius $1/(2k) \leq \epsilon$. Let C denote the collection of centers of these hypercubes, and after applying the quotient map $\pi(c) = [c]$ we obtain points $\pi(C)$ in the quotient space which are an ϵ cover of $K/\mathbb{G}$. Note that points in C which are permutations of each other are mapped by π to the same orbit. Therefore, the cardinality of the cover $\pi(C)$ is the number of equivalence classes with respect to the relation on C defined by the group: $c_1 \equiv c_2 \iff \exists \pi \in S_n : c_1 = \pi(c_2)$. As explained in more detail in the proof, the number of equivalence classes is the combinatorial expression in (4). $\square$

4.1 Hilbert Curve Canonization

Finally, we will give an intuitive explanation of the Hilbert canonization and formulate the claim that its covering number exhibits only polynomial growth in the point cloud dimensionality. For a full definition of the canonization and proof see Appendix A.5.

Figure 2: The Hilbert curves H_m for $d = 2$ and $m = 1, 2, 3$ induce an ordering on a grid in $[0, 1]^d$ with 2^{md} points. On the right we see the discontinuous lexsort ordering on the grid.



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The Hilbert curve is a space-filling curve, mapping the unit interval $[0, 1]$ continuously onto a hypercube $[0, 1]^d$. It is defined as a limit of curves $H_m : [0, 1] \rightarrow [0, 1]^d$ which are not onto but whose image becomes more and more 'dense' in $[0, 1]^d$ as m grows, as shown in Figure 2 for the case where $d = 2$. The curves H_m can be used to define an order on $[0, 1]^d$ (at least on points in the image of the curve) and thus induce a canonization. Intuitively, the advantage of this order over the

lexsort ordering is that this ordering is continuous (see Figure 2) again. However, while previous work [29, 6] cited this continuity as justification for Hilbert ordering, a formal justification of why this continuity is important for learning has not yet been provided. We address this caveat by showing the polynomial growth of the covering number of Hilbert canonizations:

Theorem 4.3. *Let $d, n, m \geq 2$ be natural numbers, let $\epsilon \in (0, 1)$, and assume that $\epsilon > 2^{-m-1}$ set $K = [0, 1]^{d \times n}$ and $\mathbb{G} = S_n$. Then the covering number of the Hilbert canonization c_m satisfies*

$$\mathcal{N}(c_m(K), \rho_\infty, \epsilon) \leq \binom{n + \lceil 1/(2\delta) \rceil - 1}{n}, \text{ where } \delta = \frac{(\epsilon - 2^{-m-1})^d}{4}$$

Proof idea. When restricting to n -tuples in K which are all in the image of H_m , the Hilbert canonization can be written as $H_m \circ \text{sort} \circ H_m^{-1}$. Since sort (Proposition 3.3) is an isometry, the image of $\text{sort} \circ H_m^{-1}$ can be covered efficiently. This cover can then be pushed forward to cover the image of $H_m \circ \text{sort} \circ H_m^{-1}$, using the fact that H_m is Holder continuous. $\square$

Theorem 4.3 shows that the covering number of the Hilbert canonization grows at worst polynomially in n , for fixed $\epsilon > 0$. This is in contrast with the exponential growth of the sorting canonization, proving a substantial gap between them.

Conclusion 3

Our analysis proves that the covering number of Hilbert canonizations is far superior to lexsrt canonization, giving a theoretical justification for its superior performance in practice.

To obtain more intuition for the gap between our bounds for the covering numbers, in Table 4.1 we present bounds for the covering number of the quotient space, the Hilbert canonization and the sorting canonization, as well as the covering number of the hypercube $[0, 1]^{d \times n}$ which is known to be k^{nd} when $\epsilon = 1/(2k)$ (see proof in A.5). We take characteristic values encountered in point cloud learning scenarios: point dimensions are $d = 3$, a very modest value of $\epsilon = 1/6$, and n varying between 250 and 2000.

Table 1: Comparison of covering number bounds as a function n between the quotient space, the Hilbert and Lexsort canonizations, and the hypercube

Covering number vs. n	$n = 250$	$n = 500$	$n = 750$	$n = 1000$	$n = 2000$
Quotient (upper bound)	2.1×10^{36}	7.4×10^{43}	2.2×10^{48}	3.5×10^{51}	2.0×10^{59}
Hilbert (upper bound)	5.3×10^{193}	7.9×10^{278}	5.0×10^{336}	5.0×10^{380}	4.4×10^{494}
Lexsort (lower bound)	1.1×10^{239}	4.0×10^{477}	1.4×10^{716}	5.2×10^{954}	9.2×10^{1908}
Hypercube (exact)	6.9×10^{357}	4.8×10^{715}	3.3×10^{1073}	2.3×10^{1431}	5.3×10^{2862}

We note that our bounds for the quotient space and Hilbert canonizations are upper bounds, while the Lexsort bound is a lower bound. Thus, the gap between the covering number of Lexsort and that of the other methods may be even larger than shown in the table. We also note that despite the conservative choice of $\epsilon = 1/6$, all covering numbers in the table are enormous. This picture may be over-pessimistic, as the 'true data distribution' will be supported on a 'small' set $K \subseteq [0, 1]^{d \times n}$ of 'realistic point clouds'. An experiment estimating the covering number on the Modelnet40 dataset will be shown later on in Table 2.

5 Experiments

We conduct several experiments to verify our theoretical insights and their manifestation in practical learning scenarios.

Covering Number on ModelNet In our first experiment, we considered the ModelNet [30] point cloud classification task. Our first goal was to see whether the covering number hierarchy between the four different methods considered in Table 4.1 is preserved when replacing the large domain $[0, 1]^{3 \times n}$ with the 'ModelNet point-cloud domain' K . Namely, our interest is in estimating the covering

number of K , its image under the lexsort and Hilbert canonizations, and the covering number of the quotient space $K/\mathbb{G}$.

We do not have access to the 'true point cloud' domain K , but only to the train and test sets. Thus, to estimate the covering number, we compute, for each test sample s , the minimum distance from s to all training samples (for computational efficiency, we consider only samples with the same label as s). We call this number q_s . We can then estimate the covering number by taking $\epsilon = \max_{s \in \text{Test}} q_s$, as this implies that the train set is an ϵ -cover of the test set. We call this measure the *maximal coverage*. We also consider a more robust measure, the *average coverage* $\text{mean}\{q_s | s \in \text{Test}\}$. The results, shown in Table 2, show that the expected covering number hierarchy is also preserved when considering real data.

Table 2: Distance from each test sample to its nearest train sample for $n = 256$ on ModelNet10 and ModelNet40, reporting both mean and max over the test set. Lower is better.

Method	Mean coverage ↓		Max coverage ↓	
	ModelNet10	ModelNet40	ModelNet10	ModelNet40
Regular distance	0.4545	0.3821	0.629	0.7304
Sort distance	0.3770	0.2988	0.5407	0.5550
Hilbert distance	0.2013	0.1955	0.5283	0.5490
Group distance	0.0917	0.0957	0.2742	0.4880

Classification on ModelNet Next, we consider whether the hierarchy in covering numbers between the different methods translates into a corresponding hierarchy in accuracy and generalization error. We choose a simple, purely non-invariant architecture: a Global MLP, and compare the performance of this network (i) without canonization, (ii) with lexsort canonization, and (iii) with Hilbert canonization (we cannot implement group averaging here since the group is very large). The results, shown in Table 3, reveal that the expected hierarchy is obtained, with Hilbert performing best both on ModelNet40 and ModelNet10, followed by the lexsort canonization and a plain MLP. The Generalization error column (defined as Test acc.-Train acc.) indicates that this performance gap is due to improved generalization.

Table 3: On the ModelNet classification task, we compare vanilla MLP architectures in three configurations: (a) no canon. (b) lexsort canon. and (c) Hilbert canon. As predicted by our theory, the Hilbert canon. leads to better generalization, which translates into improved overall accuracy.

Model	ModelNet10		ModelNet40	
	Test Acc ↑	Gen. error ↓	Test Acc ↑	Gen. error ↓
MLP (no canon.)	0.57	0.43	0.442	0.558
MLP (lexsort)	0.755	0.226	0.693	0.2556
MLP (Hilbert)	0.7815	0.1167	0.74	0.0861

To further verify the generality of our findings, we also trained the three alternatives on smaller samples from the ModelNet40 dataset. We find that the Hilbert > lexsort > no canon. hierarchy is preserved in all experiments, as shown in Table 9. We note that this experiment is only illustrative; the results attained by simple vanilla MLPs, even with Hilbert canonization, are far from state-of-the-art. In the context of our theory, this can also be explained through the fact that successful models for this task are typically permutation invariant, and hence their generalization is governed by the covering number of the quotient space, which is always better than the covering number of canonizations (Conclusion 1).

We next conduct some experiments to corroborate conclusion 1, which claims that, in terms of generalization, group averaging is preferable to canonizations, which are in turn preferable to ignoring invariance altogether.

Group averaging on PCA frames Our first experiment is still with the ModelNet40 classification task, but while the previous experiments investigates canonization w.r.t. permutations, and implicitly

relies on the fact that ModelNet models are aligned w.r.t rotations, in this experiment we use the invariant Deepsets [32] to deal with permutation invariant, and will focus on rotation invariance: we will assume the ModelNet models rotation degree of freedom is initially unknown, and is fixed using the PCA axes as suggested in [22]. Assuming the point cloud singular values are pairwise distinct, this procedure determines the point cloud orientation uniquely up to a $\{-1, 1\}^3$ symmetry.

We consider four ways to deal with this symmetry: (i) Pure PCA, where the symmetry is ignored (ii) Skewness canonization, where the global sign of the x, y and z coordinates is chosen so that their third moment is positive (iii) group averaging and (iv) group augmentation where in each epoch a random group element is applied to the input. This last example can be seen as an approximation of group averaging. The results of this experiment, detailed in Table 4 and visualized in Figure 3, demonstrate a clear performance hierarchy. Group averaging achieves the highest overall accuracy, while group augmentation and skewness canonization yield comparable final results, with all three methods significantly outperforming the pure PCA baseline. Furthermore, as highlighted in Figure 3, the group augmentation approach suffers from a noticeably slower convergence rate compared to deterministic skewness canonization.

Group averaging for image classification Lastly, we check our approach on image classification. We take the rotated MNIST [15] dataset, and consider the action of the p_4 group, which is the four-element group generated by 90 degree rotations. We consider 4 approaches: simple CNN, the canonization from [14] with random or (CN(p4 frozen)-CNN) or learned CN(p4)-CNN weights, and a group average over the 4 elements of the group. As shown in Table 5, the canonizations outperform the simple CNN, but are outperformed by group averaging, as predicted in Conclusion 1. We also see in the table that the accuracy is correlated with the covering numbers of the different methods.

Table 4: *Test accuracy on ModelNet40 with a DeepSets model with orientation determined by PCA, and sign ambiguity handled via averaging, sign augmentation, canonization, or Pure PCA (ignoring the ambiguity).*

Method	Accuracy $\uparrow$
Sign Averaging	0.760 $\pm$ 0.0025
Sign Augmentation	0.742 $\pm$ 0.0022
Skewness Canonization	0.742 $\pm$ 0.0050
Pure PCA	0.729 $\pm$ 0.0062

Table 5: *Accuracy and mean coverage on Rotated-MNIST which is invariant with respect to the group p_4 of 90 rotations. We consider a non-invariant CNN, two canonizations from [14], and group averaging.*

Model	Test Acc. $\uparrow$	Coverage $\downarrow$
CNN	94.8 $\pm$ 0.13	0.185
CN(p4)-CNN	96.47 $\pm$ 0.24	0.176
CN(p4 frozen)-CNN	95.3 $\pm$ 0.2	0.176
AvgCNN	97.3 $\pm$ 0.001	0.167

6 Conclusion

In this paper, we establish a methodology for comparing different canonizations with group averaging and other equivariant models by studying their covering number. We show the covering number of canonizations is at best the covering number of the quotient space (for isometric canonizations), and at worst the covering number of the original domain (for certain discontinuous canonizations). These results suggest that group-averaged models are preferable when computationally feasible. We also show that the covering number of Hilbert canonizations for point clouds is substantially better than that of lexsort canonizations.

Limitations and Future Work A limitation of our analysis is that it relies on upper bounds on the generalization error (Proposition 2.2), which may not be tight. Nonetheless, our experiments indicate that lower covering numbers do correlate with better generalization and learning. Looking towards future developments, a central contribution of this work is the demonstration that a rigorous theoretical comparison between competing canonization methods can be established via their covering numbers. This framework complements the traditional focus on a model’s ability to distinguish between complex symmetric examples, instead highlighting properties that arguably exert a more significant influence on the ultimate success of the learning process. In future work, we intend to apply our framework to other canonizations and extend it to enable a rigorous treatment of the generalization of frame averaging.

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406 A Proofs

407 In this section, we provide the proofs that are missing in the main text.

408 A.1 Generalization and Covering Numbers

409 Our generalization results are based on the results of [31]: a combination of Theorem 1 and Example
410 4 from [31] yields the following theorem:

411 **Theorem A.1** ([31]). *If $\mathcal{Z}$ is compact w.r.t. metric ρ , and $l(h_S, \cdot)$ is Lipschitz continuous with*
412 *Lipschitz constant $c(s)$, i.e.,*

$$|l(h_S, z_1) - l(h_S, z_2)| \leq c(S)\rho(z_1, z_2), \quad \forall z_1, z_2 \in \mathcal{Z},$$

413 *and the training sample set S is generated by n IID draws from μ , then for any $\delta, \gamma > 0$, with*
414 *probability at least $1 - \delta$ we have*

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{emp}(h_S)| \leq c(S)\gamma + M\sqrt{\frac{2\mathcal{N}(\mathcal{Z}, \rho, \gamma/2) \ln 2 + 2 \ln(1/\delta)}{n}}. \quad (5)$$

415 We now explain how Theorem 2.1 from the main text follows from Theorem A.1. We recall Theorem
416 2.1:

Theorem 2.1. *Let (X, ρ_X) and (Y, ρ_Y) be metric spaces and assume X is compact. Let $f : X \rightarrow Y$
be a c_f Lipschitz function, and let $\ell : Y \times Y \rightarrow [0, M]$ be a c_ℓ Lipschitz function. Then for any
 $\epsilon, \delta > 0$ and natural n , for a training sample set S generated by n IID draws from a distribution μ ,
with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{emp}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(X, \rho_X, \epsilon) \ln 2 + 2 \ln(1/\delta)}{n}}$$

417 where c_{h_S} denotes the Lipschitz constant of h_S .

Proof. We will apply Theorem A.1, where to apply it to the setting discussed in Theorem 2.1, we
set

$$z = (x, f(x)) \quad \mathcal{Z} = \{(x, f(x)) \mid x \in X\}, \quad \rho(z, z') = \rho_X(x, x') + \rho_Y(y, y'),$$

and

$$l(h_S, z) = \ell(h_S(x), f(x)).$$

418 The measure μ on X is pushed forward to a measure $\hat{\mu}$ on $\mathcal{Z}$ via the mapping $x \mapsto (x, f(x))$.

419 We note that l is Lipschitz with a Lipschitz constant of $c_\ell \max\{1, c_{h_S}\}$ as the following shows:

$$\begin{aligned} |l(h_S, z) - l(h_S, z')| &= |\ell(h_S(x), y) - \ell(h_S(x'), y')| \\ &\leq c_\ell (\rho_Y(h_S(x), h_S(x')) + \rho_Y(y, y')) \\ &\leq c_\ell (c_{h_S}\rho_X(x, x') + \rho_Y(y, y')) \\ &\leq c_\ell \max\{1, c_{h_S}\} (\rho_X(x, x') + \rho_Y(y, y')) \\ &\leq c_\ell \max\{1, c_{h_S}\} \rho(z, z') \end{aligned}$$

420 Next, we note that under the assumptions of Theorem 2.1, the covering number of $\mathcal{Z}$ is related to the
421 covering number of X via

$$\forall \epsilon > 0, \quad \mathcal{N}(\mathcal{Z}, \rho, (1 + c_f)\epsilon) \leq \mathcal{N}(X, \rho_X, \epsilon) \quad (6)$$

422 since we have the upper bound

$$\rho((x, f(x)), (x', f(x'))) = \rho_X(x, x') + \rho_Y(f(x), f(x')) \leq \rho_X(x, x') + c_f \rho_X(x, x')$$

423 and therefore any cover of X by balls of radius ϵ is a cover of $\mathcal{Z}$ by balls of radius $(1 + c_f)\epsilon$.

Now to conclude the proof, choose any $\epsilon > 0$ and set $\gamma = 2(1 + c_f)\epsilon$. Since l is $c_\ell \max\{1, c_{h_S}\}$ Lipschitz, we have

$$\begin{aligned} |\mathcal{L}_{exp}(h_S) - \mathcal{L}_{emp}(h_S)| &\stackrel{(5)}{\leq} c_\ell \max\{1, c_{h_S}\} \gamma + M \sqrt{\frac{2\mathcal{N}(\mathcal{Z}, \rho, \gamma/2) \ln 2 + 2 \ln(1/\delta)}{n}} \\ &= 2c_\ell \max\{1, c_{h_S}\} (1 + c_f) \epsilon + M \sqrt{\frac{2\mathcal{N}(\mathcal{Z}, \rho, (1 + c_f)\epsilon) \ln 2 + 2 \ln(1/\delta)}{n}} \\ &\stackrel{(6)}{\leq} 2c_\ell \max\{1, c_{h_S}\} (1 + c_f) \epsilon + M \sqrt{\frac{2\mathcal{N}(X, \rho_X, \epsilon) \ln 2 + 2 \ln(1/\delta)}{n}} \end{aligned}$$

which proves Theorem 2.1. $\square$

Next, we prove proposition 2.2.

Proposition 2.2. *Let $(K, \rho, \mathbb{G})$ be a module. Let (Y, ρ_Y) be a metric space, and assume K is compact. Let $f : K \rightarrow Y$ be a c_f Lipschitz function **which is $\mathbb{G}$ invariant**, and let $\ell : Y \times Y \rightarrow [0, M]$ be a c_ℓ Lipschitz function, then for any $\epsilon, \delta > 0$ and natural n , for a training sample set S generated by n IID draws from the distribution μ , with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{emp}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\} (1 + c_f) \epsilon + M \sqrt{\frac{2\mathcal{N}(h_S) \ln 2 + 2 \ln(1/\delta)}{n}}$$

where

$$\mathcal{N}(h_S) = \begin{cases} \mathcal{N}(K, \rho, \epsilon) & \text{if } h_S \text{ is } c_{h_S} \text{ Lipschitz} \\ \mathcal{N}(c(K), \rho, \epsilon) & \text{if } h_S = \tilde{h}_S \circ c, \tilde{h}_S \text{ is } c_{h_S} \text{ Lipschitz, and } c \text{ is a canon.} \\ \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) & \text{if } h_S \text{ is } c_{h_S} \text{ Lipschitz and } \mathbb{G} \text{ invariant} \end{cases} \quad (1)$$

Proof. The case where h_S is Lipschitz (but not invariant) follows directly from Theorem 2.1.

Next, we consider the case that $h_S = \tilde{h}_S \circ c$ where $\tilde{h}_S$ is c_{h_S} lipschiz. We want to consider the domain to be $c(K)$, so consider the following distribution measure on $c(K)$:

$$\tilde{\mu}(A) = \mu(c^{-1}(A))$$

Note that

$$\forall k \in c(K), h_S(k) = \tilde{h}_S(k)$$

Next, sample n IID samples from K denoted by $x_1, x_2, \dots, x_n$ and note that $c(x_1), c(x_2), \dots, c(x_n)$ are an IID samples from $c(K)$ by $\tilde{\mu}$ and denote them by $s_1, s_2, \dots, s_n$. Note that

$$\mathcal{L}_{emp}(h_S) = \frac{1}{n} \sum_{i=1}^n \ell(h_S(x_i), f(x_i)) = \frac{1}{n} \sum_{i=1}^n \ell(\tilde{h}_S(s_i), f(s_i)) = \mathcal{L}_{emp}(\tilde{h}_S)$$

So, the empirical losses are the same, and next we show the expected losses:

$$\begin{aligned} \mathcal{L}_{exp}(\tilde{h}_S) &= \mathbb{E}_{x \sim \tilde{\mu}} \ell(\tilde{h}_S(x), f(x)) = \int_{c(K)} \ell(\tilde{h}_S(x), f(x)) d\tilde{\mu} = \int_{c^{-1}(c(K))} \ell(\tilde{h}_S(c(x)), f(c(x))) d\mu \\ &= \int_K \ell(h_S(x), f(x)) d\mu = \mathcal{L}_{exp}(h_S) \end{aligned}$$

Finally, we apply Theorem 2.1 on $\tilde{h}_S$ and as it's c_{h_S} lipschiz we obtain that with probability at least δ holds that:

$$|\mathcal{L}_{exp}(\tilde{h}_S) - \mathcal{L}_{emp}(\tilde{h}_S)| \leq 2c_\ell \max\{1, c_{h_S}\} (1 + c_f) \epsilon + M \sqrt{\frac{2\mathcal{N}(c(K), \rho_X, \epsilon) \ln 2 + 2 \ln(1/\delta)}{n}}$$

And as $|\mathcal{L}_{exp}(\tilde{h}_S) - \mathcal{L}_{emp}(\tilde{h}_S)| = |\mathcal{L}_{exp}(h_S) - \mathcal{L}_{emp}(h_S)|$

We obtain that

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{emp}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\} (1 + c_f) \epsilon + M \sqrt{\frac{2\mathcal{N}(c(K), \rho_X, \epsilon) \ln 2 + 2 \ln(1/\delta)}{n}}$$

439 In the second case, where h_S is c_{h_S} lipshiz and invariant, we want to consider the domain $K/\mathbb{G}$ and
 440 define $\pi : K \rightarrow K/\mathbb{G}$ by $\pi(k) = [k]$. Define the following distribution on $K/\mathbb{G}$ by:

$$\tilde{\mu}(A) = \mu(\pi^{-1}(A))$$

Define

$$\tilde{h}_S, \tilde{f} : k/\mathbb{G} \rightarrow \mathbb{R}$$

by

$$\tilde{h}_S([x]) = h_S(x), \tilde{f}([x]) = f(x)$$

441 Note that by invariance, those functions are well defined. Next, sample n IID samples from K
 442 denoted by $x_1, x_2, \dots, x_n$ and note that $\pi(x_1), \pi(x_2), \dots, \pi(x_n)$ are an IID samples from $K/\mathbb{G}$ by
 443 $\tilde{\mu}$ and denote them by $s_1, s_2, \dots, s_n$. Note that

$$\mathcal{L}_{\text{emp}}(h_S) = \frac{1}{n} \sum_{i=1}^n \ell(h_S(x_i), f(x_i)) = \frac{1}{n} \sum_{i=1}^n \ell(\tilde{h}_S(s_i), \tilde{f}(s_i)) = \mathcal{L}_{\text{emp}}(\tilde{h}_S)$$

444 So, the empirical losses are the same, and next we show the expected losses:

$$\begin{aligned} \mathcal{L}_{\text{exp}}(\tilde{h}_S) &= \mathbb{E}_{x \sim \tilde{\mu}} \ell(\tilde{h}_S(x), \tilde{f}(x)) = \int_{K/\mathbb{G}} \ell(\tilde{h}_S(x), \tilde{f}(x)) d\tilde{\mu} = \int_{\pi^{-1}(K/\mathbb{G})} \ell(\tilde{h}_S(\pi(x)), \tilde{f}(\pi(x))) d\mu = \\ &= \int_K \ell(\tilde{h}_S([x]), \tilde{f}([x])) d\mu = \\ &= \int_K \ell(h_S(x), f(x)) d\mu = \mathcal{L}_{\text{exp}}(h_S) \end{aligned}$$

445 Note that $\tilde{f}$ is c_f lipshiz function and $\tilde{h}_S$ is c_{h_S} lipshiz function. Finally, we apply Theorem 2.1 on
 446 $\tilde{h}_S$ and $\tilde{f}$ and as it's c_{h_S} lipshiz we obtain that with probablty at least δ holds that:

$$|\mathcal{L}_{\text{exp}}(\tilde{h}_S) - \mathcal{L}_{\text{emp}}(\tilde{h}_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M \sqrt{\frac{2\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \ln 2 + 2 \ln(1/\delta)}{n}}$$

447 And as $|\mathcal{L}_{\text{exp}}(\tilde{h}_S) - \mathcal{L}_{\text{emp}}(\tilde{h}_S)| = |\mathcal{L}_{\text{exp}}(h_S) - \mathcal{L}_{\text{emp}}(h_S)|$

448 We obtain that

$$|\mathcal{L}_{\text{exp}}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M \sqrt{\frac{2\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \ln 2 + 2 \ln(1/\delta)}{n}}$$

449

□

450 A.2 Proof for the equivariant case

Proposition A.2. *Let $(K, \rho, \mathbb{G})$ be a module. Let (Y, ρ_Y) be a metric space, and assume K is compact. Let $f : K \rightarrow Y$ be a c_f Lipschitz function **which is $\mathbb{G}$ equivariant**, and let $\ell : Y \times Y \rightarrow [0, M]$ be a c_ℓ Lipschitz function and $\mathbb{G}$ invariant, then for any $\epsilon, \delta > 0$ and natural n , for a training sample set S generated by n IID draws from the distribution μ , with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{\text{exp}}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M \sqrt{\frac{2\mathcal{N}(h_S) \ln 2 + 2 \ln(1/\delta)}{n}}$$

451 where

$$\mathcal{N}(h_S) = \begin{cases} \mathcal{N}(K, \rho, \epsilon) & \text{if } h_S \text{ is } c_{h_S} \text{ Lipschitz} \\ \mathcal{N}(c(K), \rho, \epsilon) & \text{if } h_S = \tilde{h}_S \circ c, \tilde{h}_S \text{ is } c_{h_S} \text{ Lipschitz, and } c \text{ is a canon.} \\ \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) & \text{if } h_S \text{ is } c_{h_S} \text{ Lipschitz and } \mathbb{G} \text{ Equivariant} \end{cases} \quad (7)$$

452 *Proof.* We begin with the second case. Define $l(h_S, z) = \ell(h_S(x), f(x))$. We frist show it's $\mathbb{G}$
 453 invariant w.r.t z :

$$l(h_S, g \cdot z) = \ell(h_S(g \cdot x), f(g \cdot x)) = \ell(g \cdot h_S(x), f(g \cdot x))$$

454

□

455 **A.3 Proofs for Section 3**

456 We restate and prove Proposition 2.4.

457 **Proposition 2.4.** *[See proof in A.3] Let $(K, \rho, \mathbb{G})$ be a module, and assume $\mathbb{G}$ is finite. Then*

$$\mathcal{N}(K, \rho, \epsilon) \leq |\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon).$$

458 *Proof.* Let C be an ϵ -cover of $K/\mathbb{G}$ under $\rho_{\mathbb{G}}$, and define

$$\hat{C} := \{g \cdot y \mid y \in C, g \in \mathbb{G}\}.$$

459 Then of course $|\hat{C}| \leq |\mathbb{G}| \cdot |C|$.

460 We claim that $\hat{C}$ is an ϵ -cover of K under d . Let $x \in K$. Since C covers $K/\mathbb{G}$, there exists $y \in C$
461 such that

$$\rho_{\mathbb{G}}([x], [y]) \leq \epsilon.$$

462 Because $\mathbb{G}$ is finite, the minimum in the definition of $\rho_{\mathbb{G}}$ is attained, so there exists $g \in \mathbb{G}$ such that

$$\rho(x, g \cdot y) \leq \epsilon.$$

463 By construction, $g \cdot y \in \hat{C}$, hence $\hat{C}$ covers K . Therefore $\mathcal{N}(K, \rho, \epsilon) \leq |\hat{C}| \leq |\mathbb{G}| \cdot |C| =$
464 $|\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon)$. $\square$

465 We next prove Claim 3.2.

466 **Claim 3.2.** *[proof in A.3] Any isometric canonization $c : K \rightarrow K$ is (K, ϵ) optimal for all $\epsilon > 0$.*

467 *Proof.* As it always holds that

$$\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \mathcal{N}(c(K), \rho, \epsilon)$$

468 we just need to prove the opposite direction.

469 Let C be a cover of $K/\mathbb{G}$ with $\mathcal{N}(K/\mathbb{G}, \rho, \epsilon)$ elements. Consider

$$\hat{C} := \{c(x) \mid x \in C\}$$

470 and we will prove that $\hat{C}$ is an ϵ cover of $c(K)$.

471 Let $y \in c(K)$, then by definition $\exists x \in K : y = c(x)$. By the definition of cover $\exists z \in C$ such that
472 $\rho([x], [z]) \leq \epsilon$. But as $\rho_{\mathbb{G}}([x], [z]) = \rho(c(x), c(z))$ we obtain that $\rho(y, c(z)) \leq \epsilon$. So $\hat{C}$ is an ϵ cover
473 of $c(K)$ as desired. $\square$

474 **Example 3.3.** The following canonizations are isometries (see proof in A.3)

- 475 1. The canon. $c(x) = |x|$ with respect to the action of $\{-1, 1\}$ on $\mathbb{R}$ and the standard metric.
- 476 2. The canon. $\text{sort} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ with respect to the action of permutations and a metric ρ
477 induced by a p norm.
- 478 3. The centralization canon. defined for point clouds in $\mathbb{R}^{d \times n}$ with respect to the action of
479 translation by vectors in $\mathbb{R}^d$, with the metric ρ induced by the Frobenius norm.

Proof. The canonization $c(x) = |x|$ is an isometry as

$$\forall x, y \in K, \rho_{\mathbb{G}}([x], [y]) = \min\{|x - y|, |x + y|\} = ||x| - |y|| = |c(x) - c(y)|$$

480 The fact that sorting is an isometry with respect to the metric $\rho_{\mathbb{G}}$ induced from a p norm (which is the
481 p -Wasserstein metric) is shown in Lemma 4.2 in [4].

consider the action of $\mathbb{R}^d$ on $\mathbb{R}^{d \times n}$ by translating all columns of a matrix in $\mathbb{R}^{d \times n}$ by a vector in $\mathbb{R}^d$. Let ρ be the metric induced from the Frobenius norm $\rho(X, Y) = \|X - Y\|_F$, and let c be the centralization mapping $c(X) = X - \frac{1}{n}X1_{n \times n}$. This mapping is a canonization and an isometry. To see it is an isometry, note that we have for any canonization $\rho(c(X), c(Y)) \leq \rho_{\mathbb{G}}([X], [Y])$, and we

get the converse inequality by noting that c is the orthogonal projection from the inner product space $\mathbb{R}^{d \times n}$ onto the subspace $\{X \in \mathbb{R}^{d \times n} \mid X1_{n \times n} = 0\}$. Thus, let $g \in \mathbb{G}$ be a translation vector such that $\rho_{\mathbb{G}}([X], [Y]) = \rho(X, gY)$. Then

$$\rho_{\mathbb{G}}([X], [Y]) = \rho(X, gY) = \|X - gY\|_F \leq \|c(X) - c(gY)\|_F = \|c(X) - c(Y)\|_F = \rho(c(X), c(Y))$$

482

□

483 We restate and prove 3.4.

484 **Proposition 3.4.** [proof in A.3] Let $(V, \rho, \mathbb{G})$ be a module, where $\mathbb{G}$ is a finite group. Let c be a
 485 canonization such that c has $|\mathbb{G}|$ partial limits at a point $x \in V$, then for any small enough $\epsilon > 0$
 486 there exists a $\mathbb{G}$ invariant set $K \subseteq V$ such that $\mathcal{N}(c(K), \rho, \epsilon) = \mathcal{N}(K, \rho, \epsilon) = |\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho, \epsilon)$.

Proof. L is a partial limit of c if there is a sequence $x_n \rightarrow x$ such that $c(x_n) \rightarrow L$. We claim that L must be in the orbit of x . Indeed, $c(x_n) = g_n x_n$ for appropriate $g_n \in G$. Since $\mathbb{G}$ is finite, by passing to a subsequence, we can assume that $g_n = g \in G$ does not depend on n . Since $x \mapsto gx$ is an isometry, we know that $gx_n \rightarrow gx = L$. In particular, $gx \neq x$ for every $g \neq e$. Choose some $\epsilon < \min_{g \neq e} |x - g \cdot x|$, and define the G invariant set $K = K(\epsilon)$ by $K := \cup_{g \in G} B(g \cdot x, \epsilon)$. Note that $c(K)$ contains all partial limits of c at x , and therefore $gx \in c(K)$ for all $g \in G$. Since the distance between any two elements in the orbit of x is larger than ϵ , an ϵ cover of $c(K)$ must contain at least $|\mathbb{G}|$ elements. On the other hand, K itself is a union of $|\mathbb{G}|$ balls of radius ϵ , and therefore

$$\mathcal{N}(c(K), \rho, \epsilon) \leq \mathcal{N}(K, \rho, \epsilon) \leq |\mathbb{G}| \leq \mathcal{N}(c(K), \rho, \epsilon),$$

487 and therefore all these inequalities are equalities and the covering number is exactly $|\mathbb{G}|$.

488 Finally, note that $\rho(K/G, \rho_G, \epsilon) = 1$ as one ball of radius ϵ is enough for covering the quotient space,
 489 as all $|\mathbb{G}|$ balls are the same up to group action. □

490 A.4 Proofs for Section 4

491 For proving Lemma 4.2, we need the following lemma:

492 **Lemma A.3.** Let $S := [m]^n$ all vectors with n elements, each between 1 and m . Define two vectors
 493 $v \equiv u \iff \exists \pi \in S_n : u = \pi \cdot v$. Then the number of equivalent classes is exactly $\binom{n+m-1}{n}$.

494 *Proof.* Define $\forall i \in [m], c_i^u$ to be the number of times i appears in u .

495 Note that $u \equiv v \iff \forall i \in [m] : c_i^u = c_i^v$. So we want to ask how many solutions there are to the
 496 following system:

$$\begin{aligned} \sum_{i=1}^m c_i &= n \\ \forall i, c_i &\geq 0 \end{aligned}$$

497 By [24] we know the number of solutions to the equation is $\binom{n+m-1}{n}$. □

498 We now restate and prove 4.2.

499 **Lemma 4.2.** Let n, d be natural numbers and $\epsilon \in (0, 1)$. Set $K = [0, 1]^{d \times n}$ and $\mathbb{G} = S_n$. Then the
 500 covering number of the quotient space satisfies

$$\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \binom{n + k^d - 1}{n}, \text{ where } k = \left\lceil \frac{1}{2\epsilon} \right\rceil. \quad (4)$$

501 *Proof.* Set $k = \lceil 1/(2\epsilon) \rceil$. The set K can be covered by k^d hypercubes of diameter k and radius
 502 $1/(2k) \leq \epsilon$. Let C denote the collection of centers of these hypercubes, namely

$$C = \left\{ \left[\frac{2i_1 + 1}{2k}, \frac{2i_2 + 1}{2k}, \dots, \frac{2i_n + 1}{2k} \right] \mid i_j \in \{0, 1, \dots, k-1\}, 1 \leq j \leq n \right\}.$$

Applying the quotient map $\pi(c) = [c]$ we obtain points $\pi(C)$ in the quotient space which are an ϵ cover of $K/\mathbb{G}$. Note that points in C which are permutations of each other are mapped by π to the same orbit. Therefore, the cardinality of the cover $\pi(C)$ is the number of equivalence classes with respect to the relation on C defined by the group: $c_1 \equiv c_2 \iff \exists \pi \in S_n : c_1 = \pi(c_2)$. Using Lemma A.3, we know that the number of equivalence classes is

$$|\pi(C)| = \binom{n + k^d - 1}{n}$$

□

We now restate and prove 4.1.

Lemma 4.1. [proof in A.4] Let $k, d, n \geq 2$ be natural numbers, and set $\epsilon = 1/(2k)$, $K = [0, 1]^{d \times n}$ and $\mathbb{G} = S_n$. Then the covering number of the lexicographical sorting canonization satisfies

$$\left(\frac{1}{2\epsilon}\right)^{(d-1) \cdot n + 1} \leq \mathcal{N}(c_{\text{lex}}(K), \rho, \epsilon). \quad (3)$$

Proof. We consider the set $U \subset [0, 1]^{d \times n}$ defined by

$$U = \{X \in [0, 1]^{d \times n} \mid X_{11} = X_{12} = \dots = X_{1n}\}$$

The set U is not a subset of $c_{\text{lex}}(K)$, but it is a subset of its closure $\overline{c_{\text{lex}}(K)}$. This is because for every $X \in U$ we can apply an arbitrarily small perturbation of the first row of X so that it is ordered from small to large, and the X will be unaffected by the canonization. We then obtain

$$\mathcal{N}(c_{\text{lex}}(K), \rho, \epsilon) = \mathcal{N}(\overline{c_{\text{lex}}(K)}, \rho, \epsilon) \geq \mathcal{N}(U, \rho, \epsilon) \stackrel{(*)}{=} \left(\frac{1}{2\epsilon}\right)^{(d-1) \cdot n + 1},$$

where $(*)$ follows from the fact that the hypercube $[0, 1]^{(d-1) \cdot n + 1}$ can be mapped isometrically (in the ∞ norm) to U via $f : [0, 1]^{(d-1) \cdot n + 1} \rightarrow U$ by repeating the first entry $n - 1$ times, obtaining an $n \cdot d$ vector, and then transposing to be of shape $n \times d$, and from the fact that the covering number of the hypercube is $\left(\frac{1}{2\epsilon}\right)^{(d-1) \cdot n + 1}$. □

A.5 Hilbert Canonization

In this appendix, we give a formal definition of Hilbert curves and a full proof of Theorem 4.3.

The Hilbert curve $H(x)$ is defined as a limit of the function $H_m : [0, 1] \rightarrow [0, 1]^d$, and these H_m are our main focus. We will now define these curves, loosely following the terminology from [11]. Firstly, for a given integer L , we consider a partition of the unit interval into 2^L disjoint intervals: the semi-closed intervals $I_L(\ell) = [\ell 2^{-L}, (\ell + 1) 2^{-L})$ for $\ell = 0, \dots, 2^L - 2$, and the closed interval $I(2^L - 1) = [(2^L - 1) 2^{-L}, 1]$. We denote the collection of all these intervals by $\mathcal{I}_L$. For a natural $d > 1$, we also consider the partition of the unit hypercube $[0, 1]^d$ into $2^{L \cdot d}$ disjoint hypercubes of the form $E_L^d(\vec{\ell}) = I(\ell_1) \times \dots \times I(\ell_d)$. We denote the collection of these hypercubes by $\mathcal{E}_L^d$.

Next, for any given m , we define H_m to be a bijection from $\mathcal{I}_{d \cdot m}$ to $\mathcal{E}_m^d$, such that (i) adjacent intervals are mapped to adjacent hypercubes and (ii) if $I_{d \cdot (m+1)}(k') \subseteq I_{d \cdot m}(k)$ then $H_{m+1}(I_{d \cdot (m+1)}(k')) \subseteq H_m(I_{d \cdot m}(k))$. An illustration of H_m , $m = 1, 2, 3$ for $d = 2$ is given in Figure 2. Each H_m maps a finite collection of intervals bijectively to a finite collection of hypercubes. This naturally induces a bijection from the grid defined by the intervals centroids, to the grid defined by the hypercube centroids. We denote these grids by $\mathbb{A}[d \cdot m] \subseteq [0, 1]$ and $\mathbb{B}[d, m] \subseteq \mathbb{R}^d$, respectively.

The Hilbert curve H is obtained from H_m by a limiting procedure and shown to be Holder continuous [11]. Following similar ideas we show that each Holder function H_m is Holder continuous with constant that do not depend on m . This property will be crucial for the analysis of the covering number of the Hilbert canonization.

Lemma A.4. For every natural m, d , we have that

$$\forall x, y \in \mathbb{A}[d \cdot m], \quad \|H_m(x) - H_m(y)\|_\infty \leq 4|x - y|^{1/d}$$

536 *Proof.* Let $x \neq y$ be points in the one dimensional grid $\mathbb{A}[d \cdot m]$, and assume without loss of generality
 537 that $x < y$. Note that $y - x \geq \frac{1}{2^{d \cdot m}}$, since this is the minimal distance between points in the grid. On
 538 the other hand, $y - x \leq 1$. Therefore, there exists some k with $m - 1 \leq k \leq 0$, such that

$$2^{-d \cdot (k+1)} \leq |x - y| \leq 2^{-d \cdot k}. \quad (8)$$

539 It follows that x and y are in two adjacent intervals in the k -th level (or in the same interval), and
 540 by the nesting property of Hilbert's curve, we know that $H_m(x)$ and $H_m(y)$ are in two adjacent
 541 hypercubes (or in the same hypercube). Therefore, their distances is at most the sum of the diameters
 542 of those two cubes, so

$$\|H_m(x) - H_m(y)\|_\infty \leq 2 \cdot 2^{-k} \stackrel{(8)}{\leq} 4|x - y|^{1/d}$$

543 □

544 **Hilbert Canonization** As a first step, we define a canonization not on all of $[0, 1]^{d \times n}$, but rather on
 545 the finite set of n tuples of grid elements $\mathbb{B}^n[d, m]$. This is done by mapping each tuple-element to the
 546 unit interval using H_m^{-1} , then sorting the resulting n -dimensional array, and mapping back to the d -
 547 dimensional grid using H_m . We denote this canonization by c_m . We can write $c_m = H_m \circ \text{sort} \circ H_m^{-1}$,
 548 with the understanding that H_m and H_m^{-1} are applied elementwise to n -tuples of grid elements.

549 To extend H_m to all of $K = [0, 1]^{d \times n}$, we employ the following procedure: For a given $X =$
 550 $(x_1, \dots, x_n) \in K$, (i) **Rounding:** Each x_j resides in a unique hypercube in $\mathcal{E}_m^d$. Replace x_j with
 551 the centroid of this cube $y_j \in \mathbb{B}^n[d, m]$. (ii) **Sorting:** find the permutation $\tau \in S_n$ which sorts
 552 $(H_m^{-1}(y_1), \dots, H_m^{-1}(y_n))$ from small to large. (iii) **Canonizing** Apply the permutation τ to X . We
 553 note that if X contains several points in the same hypercube, the permutation τ is not uniquely
 554 defined. In this case, the order of points with the hypercube will be determined by lexicographical
 555 sorting.

556 We now restate and prove our theorem regarding the covering number of the Hilbert canonization:

Theorem 4.3. *Let $d, n, m \geq 2$ be natural numbers, let $\epsilon \in (0, 1)$, and assume that $\epsilon > 2^{-m-1}$ set
 $K = [0, 1]^{d \times n}$ and $\mathbb{G} = S_n$. Then the covering number of the Hilbert canonization c_m satisfies*

$$\mathcal{N}(c_m(K), \rho_\infty, \epsilon) \leq \binom{n + \lceil 1/(2\delta) \rceil - 1}{n}, \text{ where } \delta = \frac{(\epsilon - 2^{-m-1})^d}{4}$$

Proof. We first begin by a reduction from covering K to covering the grid $\mathbb{B}^n[d, m]$. For convenience
 we abbreviate $\mathbb{B}^n := \mathbb{B}^n[d, m]$. We note that for any $X \in K$, the matrix Y obtained by the rounding
 procedure satisfies $\|X - Y\|_\infty \leq 2^{-m-1}$. This stays true also after applying the canonization,
 $\|c_m(X) - c_m(Y)\|_\infty \leq 2^{-m-1}$, since the canonization will apply the same permutation to both X
 and Y . It follows that

$$\mathcal{N}(c_m(K), \rho_\infty, \epsilon) \leq \mathcal{N}(c_m(\mathbb{B}^n), \rho_\infty, \epsilon - 2^{-m-1}).$$

557 It remains to bound the right-hand side:

$$\begin{aligned} \mathcal{N}(c_m(\mathbb{B}^n), \rho_\infty, \epsilon - 2^{-m-1}) &= \mathcal{N}(H_m \circ \text{sort} \circ H_m^{-1}(\mathbb{B}^n), \rho_\infty, \epsilon - 2^{-m-1}) \\ &\stackrel{(*)}{\leq} \mathcal{N}(\text{sort} \circ H_m^{-1}(\mathbb{B}^n), \rho_\infty, \delta) \\ &\leq \mathcal{N}(\text{sort}([0, 1]^n), \rho_\infty, \delta) \\ &\stackrel{(**)}{\leq} \binom{n + \lceil 1/(2\delta) \rceil - 1}{n} \end{aligned}$$

558 Where $(*)$ follows from the Holder properties of H_m proven in Lemma A.4, and $(**)$ follows from
 559 the fact that one-dimensional sorting is an isometry (Proposition 3.3) and the upper bound on the
 560 covering number of $[0, 1]^n/\mathbb{G}$ (Lemma 4.2). □

561 **Lemma A.5.** *Be $K = [0, 1]^{n \times d}$ and $\epsilon = \frac{1}{2^k} > 0$ such that $k \in \mathbb{N}$. Then $\mathcal{N}(K, d_\infty, \epsilon) = \frac{1}{(2\epsilon)^{nd}}$*

562 *Proof.* It’s easy to see that each line we can cover using $\frac{1}{2\epsilon} \epsilon$ sized lines, so using the product of
563 size covering for $n \cdot d$ times we get a covering for K that uses $\frac{1}{(2\epsilon)^{nd}}$ balls. On the other hand,
564 denote by F some other covering. Note that $K \subseteq \cup_{c \in F} B(c, \epsilon)$ So $1 = \mu(K) \leq \mu(\cup_{c \in F} B(c, \epsilon)) \leq$
565 $\sum_{c \in F} \mu(B(c, \epsilon)) = |F| \cdot (2\epsilon)^{nd}$ so $\frac{1}{(2\epsilon)^{nd}} \leq |F|$. Thus, we got the lower bound and the upper
566 bound as desired. $\square$

567 B Additional Experiments

568 B.1 Laplacian Eigenvector Sign Ambiguity: Augmentation vs. Canonization on 569 ogbg-molpcba

570 This appendix presents an additional empirical comparison between deterministic canonization and
571 training-time random group augmentation, on a setting distinct from the main-text experiments
572 and complementary to the PCA experiment of Section 4. The goal is to ask whether the empirical
573 behaviour of these two strategies on a standard graph property prediction benchmark is qualitatively
574 consistent with our Proposition 2.2 and the covering-number inequality (2); we do not claim a
575 quantitative test of the inequality, since test AP on a finite-sample-trained GIN mixes optimisation,
576 capacity and generalisation effects that the inequality does not isolate. The experiment is also
577 consistent with the empirical observation cited in the Related Work that randomized SMILES can
578 outperform canonized SMILES [1, 3].

579 **Module.** We study graph property prediction on ogbg-molpcba using a 5-layer Graph Isomorphism
580 Network (GIN) with Laplacian positional encoding (LapPE). For each graph G with normalized
581 Laplacian L_G , we compute the k smallest non-trivial eigenpairs $\{(\lambda_i, v_i)\}_{i=1}^k$ and concatenate
582 the eigenvectors as a node-level positional encoding. The resulting LapPE is invariant to graph
583 isomorphism only up to the eigenvector *ambiguity group*

$$\mathbb{G}_{\text{LapPE}} = \prod_{i: m_i=1} \{-1, +1\} \times \prod_{i: m_i>1} O(m_i),$$

584 where m_i denotes the multiplicity of eigenvalue λ_i : a sign flip is admissible for each simple eigenvalue
585 and an arbitrary orthogonal change of basis is admissible within each multiplicity- m_i block. On
586 molecular graphs the spectrum is almost everywhere simple, so in practice $\mathbb{G}_{\text{LapPE}}$ reduces to
587 $\{-1, +1\}^k$ for almost all graphs. The natural questions are then (i) whether to canonize this
588 ambiguity, and (ii) whether augmentation-based approximations of group averaging behave consistent
589 with our framework.

590 **Three methods compared.** We compare three strategies for resolving $\mathbb{G}_{\text{LapPE}}$, all implemented
591 as faithful numpy ports of the reference torch code from the Laplacian canonization release of
592 [19]. The first, map (Maximal Axis Projection), resolves a simple-eigenvalue sign by forming the
593 rank-one projector $P_i = u_i u_i^\top$, grouping the coordinate indices j by the rounded column norms
594 $\|P_i[:, j]\|$ (14-decimal rounding), and walking these groups from smallest to largest norm to find
595 the first group whose indicator vector X (the sum of standard basis vectors over the group, plus
596 a uniform shift) gives $\|P_i X\| \neq 0$; the canonical sign is then $\text{sign}(u_i^\top X)$. For a multiplicity- m_i
597 block, map forms $P = U_i U_i^\top$, selects the m_i largest-norm coordinate groups, and builds the
598 canonical orthonormal basis incrementally by projecting each group-indicator vector onto the residual
599 orthogonal complement (with plain QR inside the complementary-space step). The second, oap
600 (Orthogonalized Axis Projection), differs from map in two places: the sign step rounds the column
601 norms to 6 decimals (coarser tie-grouping under near-degenerate norms), and the basis step replaces
602 the largest-norm groupings of map by a hash-based grouping of columns and uses signed-QR in the
603 complementary-space iteration. The third, random_augmented, flips the sign of each eigenvector
604 with probability 1/2 at every training-time forward pass, so the network sees a uniformly random
605 element of $\{-1, +1\}^k$ at each step. This is a finite-group augmentation that approximates group
606 averaging over $\mathbb{G}_{\text{LapPE}}$ without summing over its 2^k elements explicitly. At evaluation time the
607 random_augmented model is fed the deterministic eigsh-sign convention from the cache (no fresh
608 random draws), yielding a single deterministic prediction per graph; this is the number reported in
609 the random_augmented column of Table 6. Note that this is an *adversarial* test-time deployment of
610 an averaging-trained model (it scores a single point of the orbit rather than the orbit mean); the

611 AUG- K column reports the alternative Reynolds-style sample of K orbit elements, and we discuss
 612 the gap between the two below.

613 **Test-time averaging AUG- K .** For random_augmented models we additionally evaluate a test-time
 614 *Reynolds-style* averaged metric. At evaluation, for each test graph we draw K independent group
 615 elements $g_1, \dots, g_K \in \mathbb{G}_{\text{LapPE}}$ (each g_j a sign-flip pattern on the simple-eigenvalue subspace and
 616 a Haar-random orthogonal matrix on each multiplicity- m block); we forward-pass the model on
 617 each sign-flipped LapPE; we average the predicted per-task probabilities; and we score the averaged
 618 predictions with the official OGB AP evaluator. We use $K=8$ at $k=3$ (which approximately covers
 619 $\{-1, +1\}^3$ under uniform i.i.d. sampling: 8 draws cover $\sim 65\%$ of the 8-element group on average)
 620 and $K=16$ at $k \in \{8, 16\}$ (sparse Monte-Carlo over $\{-1, +1\}^8$ and $\{-1, +1\}^{16}$ respectively). On
 621 ogbg-molpcba the spectrum is almost everywhere simple, so the multiplicity-block factor $\prod_i O(m_i)$
 622 is degenerate in practice and the g_j samples reduce to sign-flip patterns. This is the empirical
 623 group-averaging operator of Section 2, restricted to a finite sample.

624 **Hyperparameters and protocol.** GIN, 5 layers, hidden dim $h \in \{16, 128, 512\}$,
 625 dropout 0.5, batch size 32, optimizer Adam with cosine learning-rate schedule and ini-
 626 tial rate 10^{-3} , 3 seeds per cell ($\{0, 1, 2\}$), eigval-scaling off, eigenvector cache size 15.
 627 Best-validation-AP checkpoint selection. Software stack: PyTorch 2.10, CUDA 12.8,
 628 torch_geometric/torch_scatter per the environment file; OGB evaluator version per
 629 ogb’s default Evaluator(name=‘ogbg-molpcba’). Per-arm seeding via torch.manual_seed,
 630 numpy.random.seed and random.seed; cudnn.deterministic not enforced (so results are not
 631 bit-reproducible across hardware). Standard deviations reported below are sample standard deviations
 632 (ddof=1). Code, raw results.json files, and reproduction commands are released at the project
 633 repository.

634 **Compute budget.** At $k \in \{3, 8\}$ all three arms are trained for 200 epochs with early-stop patience
 635 15. At $k=16$, where validation AP for random_augmented continues to improve well past 200
 636 epochs, all three arms are trained for 500 epochs with early stopping disabled (patience 999). Tables 6
 637 and 7 therefore report a matched-compute comparison at every (k, h) cell. Both schedules are
 638 released in the project repository.

639 **Results.** Table 6 reports mean test AP at all nine (k, h) cells for the three methods, alongside the
 640 AUG- K test-time-averaged metric for random_augmented. Table 7 zooms in on the $k=16$ slice
 641 with sample standard deviations.

Table 6: Mean test AP on ogbg-molpcba at matched compute, 3 seeds per cell ($k \in \{3, 8\}$ at 200 epochs / patience 15, $k=16$ at 500 epochs / patience 999 for all three arms). Per-cell row-best margins are mostly within seed std and none survives a Bonferroni correction across the nine cells at $\alpha=0.05$ (see Table 7 for the $k=16$ Welch p -values); the bolding therefore marks the nominal row-best mean, not a significance claim. The AUG- K column reports test-time averaging over K ambiguity-group draws applied to the random_augmented-trained model (see text).

k	h	map $\uparrow$	oap $\uparrow$	random_augmented $\uparrow$	AUG- K
3	16	0.0626	0.0641	0.0662	0.0663 ($K=8$)
3	128	0.1704	0.1682	0.1762	0.1765 ($K=8$)
3	512	0.2397	0.2431	0.2395	0.2398 ($K=8$)
8	16	0.0622	0.0605	0.0636	0.0637 ($K=16$)
8	128	0.1705	0.1754	0.1766	0.1766 ($K=16$)
8	512	0.2344	0.2371	0.2395	0.2400 ($K=16$)
16	16	0.0716	0.0703	0.0691	0.0693 ($K=16$)
16	128	0.1923	0.1836	0.1916	0.1921 ($K=16$)
16	512	0.2468	0.2419	0.2483	0.2481 ($K=16$)

Table 7: Matched-compute $k = 16$ slice of Table 6: all three arms trained for 500 epochs with patience 999. Reported are mean $\pm$ sample standard deviation over 3 seeds, plus the two-sided Welch t -test p -value for the row-best mean against the row-second-best mean. None of the three row-best margins reaches $p < 0.05$, and after a Bonferroni correction across the nine cells in Table 6 ($\alpha = 0.05/9 \approx 0.0056$) none of the row-best margins anywhere in the appendix is significant.

h	map	oap	random_augmented	Welch p (best vs. 2nd)
16	0.0716 $\pm$ 0.0067	0.0703 $\pm$ 0.0070	0.0691 $\pm$ 0.0008	≈ 0.83
128	0.1923 $\pm$ 0.0005	0.1836 $\pm$ 0.0025	0.1916 $\pm$ 0.0009	≈ 0.32
512	0.2468 $\pm$ 0.0021	0.2419 $\pm$ 0.0028	0.2483 $\pm$ 0.0026	≈ 0.48

Test-time averaging Δ is small. Across all nine (k, h) cells the difference between the single-pass deterministic-eigsh evaluation of the random_augmented model and its AUG- K averaged evaluation (Table 6, right-most two columns) lies in $[-0.0002, +0.0005]$ AP, within the seed noise at every cell. The AUG- K Δ is an aggregate metric, not a direct measurement of model invariance; we test invariance directly in the next paragraph.

Per-graph orbit stability. For each test graph we draw $K = 64$ i.i.d. elements g_j of the sign-flip subgroup of $\mathbb{G}_{\text{LapPE}}$, forward-pass the trained random_augmented model on each $g_j(\text{LapPE})$, and record the per-task predicted probability. The resulting per-graph spread (max – min predicted probability across the $K = 64$ draws) is a direct measure of how much the model’s output varies across orbit elements. Aggregating across all 43,793 ogbg-molpcba test graphs and 128 tasks, at hidden dim $h = 128$ we observe a clean monotone scaling with k (Table 8): the per-graph spread is 0.0018 at $k = 3$, 0.0034 at $k = 8$, and 0.0048 ± 0.0008 at $k = 16$ (mean over 3 seeds at $k = 16$; single seed at $k \in \{3, 8\}$). This is qualitatively consistent with $|\mathbb{G}_{\text{LapPE}}| = 2^k$: more orbit elements give more room to vary, and the model is closer to invariant for smaller k . Concretely at $k = 16/h = 128$, the per-draw test AP across $K = 64$ orbit elements has standard deviation 0.0006 within a seed and total range ≈ 0.0024 , comparable to the seed-to-seed std of single-pass evaluation in Table 7 (0.0009). The Reynolds-style (mean-probability) AP across the $K = 64$ draws is 0.1923 ± 0.0017 over seeds, within seed noise of the matched-compute single-pass number (0.1916 ± 0.0009 for random_augmented and 0.1923 ± 0.0005 for the matched-compute map arm at the same cell). We read this as direct evidence that the random_augmented model is approximately $\mathbb{G}_{\text{LapPE}}$ -invariant on the test distribution at the configurations tested.

Table 8: Per-graph orbit-stability of the trained random_augmented model under $K = 64$ uniform i.i.d. draws from the sign-flip subgroup of $\mathbb{G}_{\text{LapPE}}$, evaluated on the ogbg-molpcba test set. “Per-graph spread” is the mean over graphs and tasks of (max – min) predicted probability across the $K = 64$ orbit elements; “per-draw AP std (within seed)” is the standard deviation of the OGB AP scored on each single orbit draw. The $k = 16$ row is the mean $\pm$ sample std over the 3 training seeds; the $k \in \{3, 8\}$ rows are single-seed estimates ($s = 0$). Per-graph spread scales monotonically with $|\mathbb{G}_{\text{LapPE}}| = 2^k$.

k	h	seeds	per-graph spread	per-draw AP std (within seed)	Reynolds AP
3	128	1	0.0018	0.0004	0.1741
8	128	1	0.0034	0.0005	0.1806
16	128	3	0.0048 ± 0.0008	0.0006	0.1923 ± 0.0017

Reading the tables. Reading along the hidden-dim axis at fixed k in Table 6, the row-best AP grows monotonically with h at every k , and the row-best arm is most often random_augmented (six of nine cells), with map nominally row-best at $k = 16/h \in \{16, 128\}$ and oap nominally row-best at $k = 3/h = 512$. The matched-compute $k = 16$ slice in Table 7 resolves these margins against seed variance: the smallest sample std on each row is consistently the random_augmented std (0.0008, 0.0009, 0.0026), while map carries the largest std at small h (0.0067 at $h = 16$). Under a two-sided Welch t -test for unequal variances, the row-best versus row-second-best margins at $k = 16$ have

670 p -values of 0.83, 0.32, and 0.48 at $h = 16, 128, 512$ respectively, none significant; this is the honest
671 reading of the appendix at $n = 3$ seeds. At $k \in \{3, 8\}$ the row-best margins are similar in magnitude
672 or smaller, with `random_augmented` winning five of six rows there but at margins of the same order
673 as the seed std. We therefore present Table 6 as a nominal-rank summary, not as a benchmark claim.

674 **Tie to the theory.** On `ogbg-molpcba` at matched compute, with 3 seeds per cell, we did not
675 detect a statistically significant difference between training-time random sign-flip augmentation
676 and projection-based deterministic canonization at any of the nine (k, h) cells; this is qualitatively
677 consistent with (2) of Proposition 2.2, which suggests augmentation can match canonization on tasks
678 where the covering number of $\mathbb{G}$ is small, but it is not a positive test of the inequality, since AP gaps
679 of order 10^{-3} are dominated by optimisation and seed noise rather than by the input-space covering
680 complexity the inequality bounds. The `AUG-K` column moves AP by at most $+0.0005$ across the
681 nine cells; this is consistent with the cached deterministic eigsh-sign PE already lying close to the
682 orbit-mean the network was trained on, but is not itself a direct test of model invariance.

683 C Experimental Setup and Hyperparameter Details

684 C.1 Experimental Setup for the covering number experiment

We applied the following preprocessing for *each* sample: we first sample 256 points, then we shift it to the positive axis, and then divide by the maximum axis along x, y, z , obtaining all entries to be positive and between 0 and 1. The metric we considered on the point cloud space is

$$\rho_{\mathbb{G}}(X, Y) = \frac{1}{n} \sum_{i=1}^n \|X_i - Y_i\|_2.$$

685 The same metric is used for the two canonizations, and its quotient $\rho_{\mathbb{G}}$ is used for the quotient space.

686 C.2 Experiment setup: ModelNet40 classification

Table 9: Test accuracy of the Global MLP under varying training set sizes. The performance gap between Hilbert and lexicographical sorting widens as data becomes scarcer, highlighting the regularizing effect of optimal canonization.

Training Samples	None $\uparrow$	Lexicographical $\uparrow$	Hilbert $\uparrow$
9840 (full)	0.433	0.700	0.748
4920	0.408	0.641	0.714
2460	0.343	0.573	0.673
1230	0.307	0.504	0.601

687 In the ModelNet40 classification tasks, we sampled from each input CAD model point cloud, which
688 consists of 1024 points, and applied an initial batched affine normalization to $[0, 1]^3$. The data is then
689 ordered using one of three strategies: no canonization (baseline), lexicographical sorting (i.e., sorting
690 points sequentially by their x , then y , then z coordinates), or Hilbert curve sorting (using a space
691 size of 2^{12}). Following ordering, the coordinates are projected into a higher-dimensional space using
692 Random Fourier Features to capture high-frequency geometric details. The feature vector is then
693 passed through a bottlenecked MLP consisting of residual blocks with dimensions $[256, 128, 64]$,
694 yielding the final classification logits. The MLPs were trained for 100 epochs with a fixed batch size
695 of 256, and were trained using AdamW.

696 To ensure a rigorous and fair comparison between the different ordering strategies, we conducted
697 comprehensive grid sweeps to identify the optimal hyperparameters for the uncanonized baseline
698 (`ply`), lexicographical sorting (`lex`), and Hilbert curve sorting (`hilbert`). The specific optimal
699 hyperparameters extracted from our sweeps and utilized for the final evaluation are detailed in
700 Table 10. The best hyperparameters for each model on ModelNet40 are used for training the models
701 on ModelNet10.

Table 10: Optimal hyperparameters identified via grid search for each ordering strategy.

Method	Learning Rate	Dropout	Weight Decay	Fourier Scale	Label Smoothing	Bands
No canon.	1.63×10^{-3}	0.3	0.05	1.0	0.1	2
Lexsort	6.41×10^{-4}	0.1	0.05	0.1	0.2	3
Hilbert	9.61×10^{-4}	0.3	0.01	0.1	0.2	3

Robustness Across Seeds To account for variance in weight initialization and optimization dynamics, we evaluated the final optimal configurations across 5 independent random seeds. The averaged test accuracy and standard deviations are reported in Table 11. The results demonstrate that the Hilbert curve canonization not only achieves the highest average accuracy but also exhibits the lowest variance across different initializations.

Table 11: Test accuracy across 5 independent random seeds using the optimal hyperparameters. Hilbert canonization yields both higher performance and greater stability.

Method	Average Accuracy $\uparrow$	Standard Deviation
No canon.	0.433	0.0249
Lexsort	0.700	0.01358
Hilbert	0.748	0.00944

C.3 Experimental Setup: Modelnet 40 Data scarcity experiment

To simulate data scarcity, we subsample the ModelNet40 training set using strides of 1, 2, 4, and 8, resulting in training subsets of 9840, 4920, 2460, and 1230 samples, respectively. Results are shown in Table 9.

C.4 Experimental Setup: PCA group averaging

We use a DeepSets architecture that is fully invariant to point order permutations but strictly non-invariant to the $\{-1, 1\}^3$ sign flips. We intentionally employ a lightweight, low-capacity version of this architecture. A highly over-parameterized model might achieve higher absolute accuracies, potentially masking the generalization gaps we aim to observe; by constraining the network capacity, we ensure that any performance differences are strictly attributable to the chosen alignment strategy. Each input point cloud consists of 1024 points from the ModelNet40 dataset. We compare four approaches: (1) **Pure PCA**, which leaves the sign ambiguity unresolved (acting as a baseline or potentially *poor* canonization due to eigensolver discontinuities); (2) **Skewness**, a deterministic canonization mapping $c : K \rightarrow K$ that resolves the ambiguity by orienting axes based on the third moment; (3) **Frame Averaging**, an explicitly $\mathbb{G}$ -invariant model constructed by pooling the network’s predictions across all 8 elements of the group using log-sum-exp; and (4) **Random Frame**, which applies elements of $\mathbb{G}$ as random data augmentation during training. All models are optimized using Adam for 1600 epochs and evaluated across 5 different random seeds.

C.5 Experimental Setup: Rotated MNIST

In the last set of the experiments, we compare the performance of several image rotation invariant models. We considered three models: simple CNN, CN(p4)-CNN, and AvgCNN. The backbone for all three models is the same and additional layers are added for CN(p4)-CNN, AvgCNN. In order to have the same setting, including the data split we rerun all models, including CN(p4)-CNN. We used learning rate of $1e^{-3}$ and a batch size of 256 and an AdamW optimizer. Each experiment is run for 5 different seeds, and mean and standard deviation are reported.

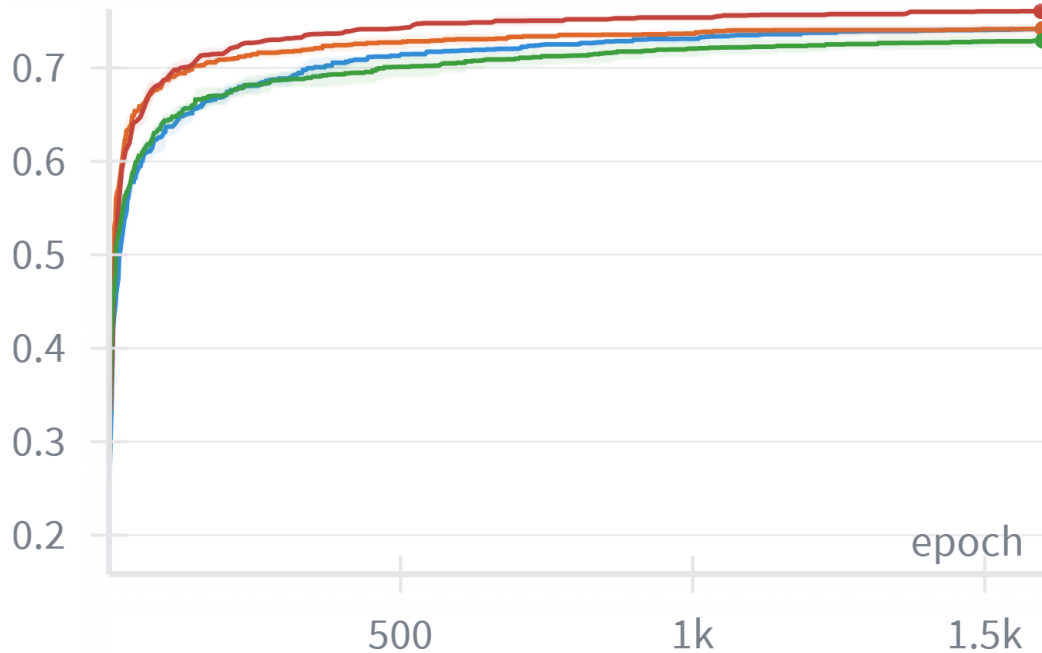


Figure 3: **Rotation Symmetry Accuracy.** The curves illustrate the comparative performance of the four approaches: Random Frame (blue), Frame Averaging (red), Skewness (orange), and the Pure PCA baseline (green). Frame Averaging achieves the strongest overall performance. While Random Frame and Skewness reach comparable final results—both significantly outperforming the Pure PCA baseline—the augmentation-based Random Frame method exhibits a noticeably slower convergence rate than deterministic canonization via Skewness.

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